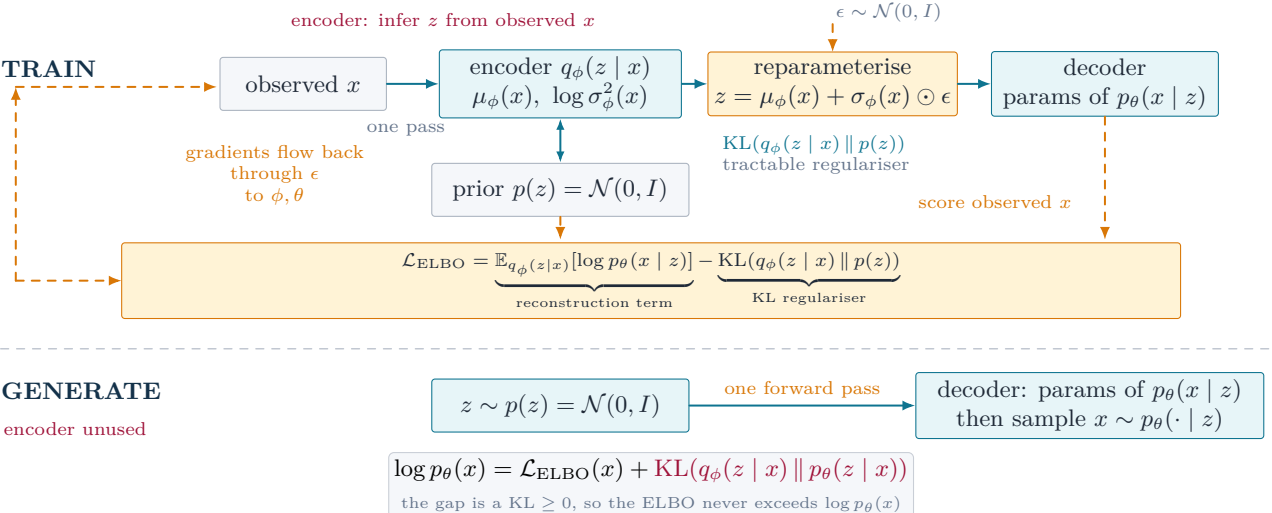
encoder: infer z from observed x
$\epsilon \sim \mathcal{N}(0, I)$
TRAIN
observed x
encoder $q_\phi(z \mid x)$
$\mu_\phi(x),\ \log \sigma^2_\phi(x)$
reparameterise
$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$
decoder
params of $p_\theta(x \mid z)$
one pass
gradients flow back
through ϵ
to ϕ, θ
prior $p(z) = \mathcal{N}(0, I)$
$\mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z))$
tractable regulariser
score observed x
$\mathcal{L}_{\mathrm{ELBO}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z))$
reconstruction term
KL regulariser
GENERATE
encoder unused
$z \sim p(z) = \mathcal{N}(0, I)$
one forward pass
decoder: params of $p_\theta(x \mid z)$
then sample $x \sim p_\theta(\cdot \mid z)$
$\log p_\theta(x) = \mathcal{L}_{\mathrm{ELBO}}(x) + \mathrm{KL}(q_\phi(z \mid x) \,\|\, p_\theta(z \mid x))$
the gap is a KL ≥ 0, so the ELBO never exceeds $\log p_\theta(x)$